

F36 (F24 v0.2) — Substrate-color² source: the amplitude-vs-rate diagnosis + share-an-operator discipline on the 45 cross-link

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F36 (F24 v0.2) Substrate-Color² Source — Diagnosis

0. Goal and scope

Advance the substrate-color² open question (the N_c^2 numerator in $\sin \theta_C$) at depth. Tracked-open, not a Saturday-close per Casey Action Item 4. v0.1 (F24) enumerated three candidate sources for $N_c^2=9$ and found none clean. This v0.2 does something more useful than a fourth candidate: it diagnoses *why* the search has stalled and reframes the open question into a sharper, falsifiable one. AC(0): count the color powers first.

1. The identification is essentially exact (positive sharpening)

$$\sin \theta_C \stackrel{?}{=} \frac{N_c^2}{2^{N_c} \cdot n_C} = \frac{9}{40} = 0.22500$$

The Friday notes compared this to $|V_{us}|$ from kaon decays (0.2243, “0.31%”). The cleaner comparison is the Wolfenstein parameter $\lambda = 0.22500$ (PDG), and against it the form is exact to all five digits (0.000%). So the *identification* is excellent — better than previously stated. The problem was never the precision. It is the mechanism for N_c^2 sitting in an amplitude.

2. The real blocker: N_c^2 in an amplitude (color counting)

$\sin \theta_C$ is an amplitude (a CKM matrix element, $|V_{us}|/\lambda$), not a rate. In the substrate reading it is an off-diagonal matrix element between generation K-types,

$$\sin \theta_C \sim \langle V_{\text{gen-1}} \otimes \text{color} \mid T_W \mid V_{\text{gen-2}} \otimes \text{color} \rangle.$$

The Standard Model weak current is color-diagonal ($T_W = I$ on color). A color-diagonal amplitude carries exactly one color trace — $\text{Tr}(I_3) = N_c$ — when summed over the color index. To get N_c^2 you need two independent color sums, which is the signature of a rate ($|\text{amplitude}|^2$, two color loops) or of a genuinely two-endpoint color structure, not of a single tree-level color-singlet amplitude.

So the N_c^2 numerator is anomalous *as an amplitude factor*. This is the precise blocker, and it is sharper and more actionable than the v0.1 framing (“ $9 = N_c^2 = (\text{rank}+1)^2 = 2^{\{N_c\}+1} = C_2 + N_c$, four ways”). The Casey #5 integer-multiplicity of 9 is a symptom; the disease is the amplitude/color-trace mismatch.

3. Two honest resolution directions (both multi-week, both need the explicit matrix element)

- Direction A — the forced object is the rate. If the substrate forces λ^2 (a rate) rather than λ (an amplitude), then N_c^2 is the natural two-color-trace factor of $|\text{color-singlet amplitude}|^2$, and $\lambda = \sqrt{(\text{rate})}$ inherits N_c^2 . Test: is the substrate’s primitive observable in the quark-mixing sector the *probability* $\sin^2 \theta_C = (9/40)^2 = 81/1600 = 0.050625$ ($= \lambda^2$, exact), with $\sin \theta_C$ derived as its root? If yes, the color² is forced by the rate’s two color sums. This is the cleaner of the two and reframes the whole CKM-sector search toward rates.
- Direction B — two-endpoint color sum. If the gen-1 and gen-2 K-types each carry an *independent* color sum in the Bergman inner product (initial-quark color $\otimes$ final-quark color, not tied by a single δ), the amplitude carries $N_c \times N_c$. This is v0.1 candidate (c). It requires showing the substrate matrix element does not reduce to a single color trace — which the explicit FK Bergman color-tensored matrix element must decide. v0.1 found the Sym^k cross-index gives 3, not 9, so Direction B is not yet supported and must not be asserted.

Neither is closed. The point of v0.2 is that the search is now two sharp falsifiable questions (is the primitive a rate? does the matrix element carry one color trace or two?) instead of an open-ended hunt for a factor of 9.

4. Discipline flag — the $N_c^2 \cdot n_C = 45$ PMNS $\leftrightarrow$ CKM link is a LEAD, not a candidate

The Saturday board has Grace adding $N_c^2 \cdot n_C = 45$ as the 6th G14 candidate (“one substrate primitive crossing quark + lepton mixing sectors”), citing PMNS $\sin^2 \theta_{13} = 1/45$ and CKM $\sin \theta_C \propto N_c^2 / (2^{\{N_c\}} \cdot n_C)$.

Apply the same discipline Cal applied to the 28-cross-link and I applied to the 81/40 vacuum factor (F35): share an operator, not an integer.

- In PMNS, N_c^2 sits in the denominator ($1/45 = 1/(N_c^2 \cdot n_C)$).
- In CKM, N_c^2 sits in the numerator ($9/40$).

- The “shared primitive” is the integer N_c^2 appearing on opposite sides of two different observables — a shared *number*, with no shared *operator* yet exhibited.

Per the standing discipline this is a LEAD, not a Schur-generator candidate. It becomes a candidate only if a single substrate operator is shown to produce N_c^2 in *both* sectors (e.g., the same color-tensored mixing operator, with the rate/amplitude distinction explaining why it lands in numerator vs denominator). Grace — recommend logging 45 as a LEAD in G14 v0.3, not a candidate, pending that operator. This keeps G14 honest and consistent with the 28-link and 81/40 calls.

5. What v0.2 closes and what stays open

Closed/sharpened by this note: - Identification quality corrected upward: $9/40 = \lambda$ exactly (0.000%), not 0.31%. - The OPEN reframed from “which of 4 ways to make 9” → the amplitude/color-trace tension (N_c^2 in an amplitude needs two color sums). - Two falsifiable directions named (A: rate-primitive; B: two-endpoint color sum). - The 45 PMNS↔CKM link downgraded to LEAD (integer-shared) per share-an-operator discipline.

Open (multi-week, Cal #189): - Compute the explicit FK Bergman color-tensored mixing matrix element; count its color traces (decides A vs B). - Determine whether the substrate’s CKM-sector primitive is the rate $\sin^2\theta_C$ or the amplitude $\sin\theta_C$. - Promote 45 from LEAD to candidate only on a shared operator.

6. Closure

F36 (F24 v0.2): the substrate-color² source for $\sin\theta_C$ is reframed. The form $9/40 = \lambda$ is an essentially exact identification; its forcing is blocked by N_c^2 sitting in an amplitude, where a color-diagonal current gives only one color trace (N_c). The productive next questions are sharp and falsifiable — is the primitive a rate (Direction A, color² natural as a two-trace) or does the matrix element carry two independent color sums (Direction B). The $N_c^2 \cdot n_C = 45$ PMNS↔CKM cross-link is a LEAD (shared integer), not a Schur candidate, until one operator produces N_c^2 in both sectors — flagged to Grace before G14 v0.3.

Tier: F36 (F24 v0.2) diagnostic forward step; substrate-color² source OPEN reframed to amplitude-vs-rate; multi-week per Cal #189; tracked-open, NOT Saturday-close.

— Lyra, Sat 2026-06-06 11:55 EDT. F36 (F24 v0.2): $9/40 = \text{Wolfenstein } \lambda$ exactly (0.000%, sharpened from Friday’s 0.31% vs kaon $|V_{us}|$); OPEN reframed from Casey-#5 multiplicity of 9 to the amplitude/color-trace tension (N_c^2 in an amplitude needs two color sums; color-diagonal current gives one trace = N_c); two falsifiable directions — A rate-primitive $\sin^2\theta_C$ (color² natural as two-trace), B two-endpoint color sum (needs explicit FK matrix element, v0.1 Sym^k gave 3 not 9 so unsupported); $N_c^2 \cdot n_C = 45$ PMNS↔CKM downgraded to LEAD (shared

integer not shared operator) per share-an-operator discipline — flag to Grace for G14 v0.3; multi-week, tracked-open.